

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 7

Forecasting and Model Selection

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MSFE · Diebold–Mariano Test · Forecast Taxonomy · Intercept Correction · GETS ·
Autometrics · IIS · LASSO · Forecast Combination · Machine Learning

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1. Forecast Evaluation

1.1. The MSFE optimality criterion

For h -step-ahead forecast $\tilde{Y}_{T+h|T}$ the mean square forecast error (MSFE) is:

$$\text{MSFE}(h) = \frac{1}{H} \sum_{t=1}^H (Y_{T+t+h} - \tilde{Y}_{T+t+h|T+t})^2.$$

In a stationary environment the conditional expectation $\hat{Y}_{T+h|T} = E[Y_{T+h} | \mathcal{I}_T]$ is the unique MSFE-minimising forecast:

$$E\{(\tilde{Y}_{T+h|T} - Y_{T+h})^2 | \mathcal{I}_T\} \geq E\{(\hat{Y}_{T+h|T} - Y_{T+h})^2 | \mathcal{I}_T\}.$$

1.2. Common metrics

Metric	Formula	Notes
RMSFE	$\sqrt{\text{MSFE}}$	Scale-dependent
MAE	$\frac{1}{H} \sum e_{t+h} $	Robust to outliers
MAPE	$\frac{1}{H} \sum e_{t+h} / Y_{t+h} $	Scale-free; undefined if $Y = 0$
Theil U	$\text{RMSFE} / \text{RMSFE}_{\text{naive}}$	< 1 : beats random walk

MSE decomposition (Theil, 1961):

$$\text{MSE} = \underbrace{(\bar{\tilde{Y}} - \bar{Y})^2}_{\text{bias}} + \underbrace{(s_{\tilde{Y}} - s_Y)^2}_{\text{variance}} + \underbrace{2(1 - \rho)s_{\tilde{Y}}s_Y}_{\text{covariance}}.$$

A well-specified model places most MSE in the covariance component (irreducible uncertainty). Large bias signals a systematic offset — intercept correction is warranted.

2. The Diebold–Mariano Test

Definition 2.1. Compare predictive accuracy of models A and B . Define the loss differential:

$$d_t = L(e_t^A) - L(e_t^B).$$

H_0 : equal accuracy, $E[d_t] = 0$. Test statistic: $DM = \bar{d} / \sqrt{\hat{V}(\bar{d})} \stackrel{a}{\sim} N(0, 1)$, where $\hat{V}(\bar{d})$ uses HAC (Newey–West) standard error.

At horizons $h > 1$, loss differentials are serially correlated (overlapping errors): use HAC bandwidth $\geq h - 1$. Harvey, Leybourne & Newbold (1997) recommend using the $t(H - 1)$ distribution for finite-sample accuracy.

3. Forecast Failure and the Taxonomy of Errors

Clements & Hendry (1999) classify forecast errors by source:

- (i) **Innovation variance** σ^2 : irreducible.
- (ii) **Estimation error** $O(T^{-1/2})$: shrinks with sample size.
- (iii) **Model misspecification**: systematic bias.
- (iv) **Location shift at the forecast origin**: a change in $\mu_y = \mu/(1 - \alpha)$ at time T produces persistent forecast errors. This is the **primary** cause of large forecast failures.

Simulations in Clements & Hendry (1999) show that location shifts at the forecast origin produce forecast errors an order of magnitude larger than other sources for medium forecast horizons. A change in α that leaves μ_y unchanged causes only small MSFE increases.

4. Robust Forecasting Methods

4.1. Intercept correction

If the last residual $\hat{\varepsilon}_T = Y_T - \hat{Y}_T$ signals a break, add it to all future forecasts:

$$\tilde{Y}_{T+1|T}^{\text{ic}} = \underbrace{\hat{\alpha}Y_T + \hat{\mu}}_{\tilde{Y}_{T+1}} + \underbrace{(Y_T - \hat{\mu} - \hat{\alpha}Y_{T-1})}_{\hat{\varepsilon}_T} = Y_T + \hat{\alpha}(Y_T - Y_{T-1}).$$

4.2. Differencing the estimated model

Difference the estimated model to remove the level:

$$\Delta \tilde{Y}_{T+1|T}^{\text{diff}} = \hat{\alpha} \Delta Y_T, \quad \tilde{Y}_{T+1|T}^{\text{diff}} = Y_T + \hat{\alpha} \Delta Y_T = \tilde{Y}_{T+1|T}^{\text{ic}}.$$

Proposition 4.1. *The 1-step intercept-corrected forecast equals the 1-step forecast from the differenced model. Both are level-robust: they start from the current level Y_T rather than the modelled mean, adapting immediately to level shifts.*

4.3. Forecast combination

Theorem 4.2 (Bates & Granger, 1969). *The equal-weight combination $\tilde{Y}^{\text{comb}} = K^{-1} \sum_{k=1}^K \tilde{Y}^{(k)}$ typically outperforms individual component models in MSFE, particularly when the components make structurally different errors.*

Optimal combination weights are obtained by OLS regression of realised values on component forecasts. In practice, equal weights often perform nearly as well (Makridakis & Winkler, 1983) and are more robust out of sample.

5. General-to-Specific Model Selection

5.1. The General Unrestricted Model

Start from a large well-specified model with N candidate regressors (GUM). A sub-model is **terminal** if: (i) the LR test against the GUM is accepted; (ii) each retained variable is individually significant.

5.2. Autometrics and multipath search

Hoover & Perez (1999) introduced multipath search: for each of the 10 least significant regressors, eliminate it and run backward elimination — yielding 10 paths. Autometrics (Doornik, 2009; Hendry & Doornik, 2014) extends this to include IIS and SIS, handling $N > T$ via block-splitting.

Proposition 5.1. *In Autometrics with target significance level α :*

- (i) The **gauge** (fraction of irrelevant variables falsely retained) converges asymptotically to α .
- (ii) The **potency** (fraction of relevant variables correctly retained) increases with the signal-to-noise ratio β_j/σ .
- (iii) The algorithm achieves gauge $\approx \alpha$ even when $N > T$ (IIS case).

6. Penalised Methods: LASSO and Ridge

Definition 6.1. *The **LASSO** (Tibshirani, 1996) solves:*

$$\hat{\beta}^{lasso} = \arg \min_{\beta} \left\{ \sum_{t=1}^T (Y_t - \mathbf{x}'_t \beta)^2 + \lambda \sum_{j=1}^N |\beta_j| \right\}.$$

The L_1 penalty drives some $\hat{\beta}_j$ exactly to zero, producing automatic variable selection.

Choose λ by time-series cross-validation (no data leakage from future observations). The elastic net combines L_1 and L_2 penalties. LASSO produces biased estimators; inference after selection is non-trivial. GETS, by contrast, controls gauge with valid t -tests and retains unbiased OLS estimates of selected coefficients.

7. Machine Learning and Forecast Combination

M4 competition (Makridakis et al., 2018): 100,000 series, 9 horizons, 61 methods.

- Equal-weight combination of ETS and ARIMA ranked highly.
- Pure neural networks underperformed classical methods on most series.
- Best overall: hybrid ML + ETS/ARIMA methods.

Classical time series methods embody decades of understanding about series dynamics and remain competitive especially at shorter horizons and smaller samples. Use ML methods to complement, not replace, classical methods. Combination of diverse approaches is usually more robust than any single model.

8. Core References

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